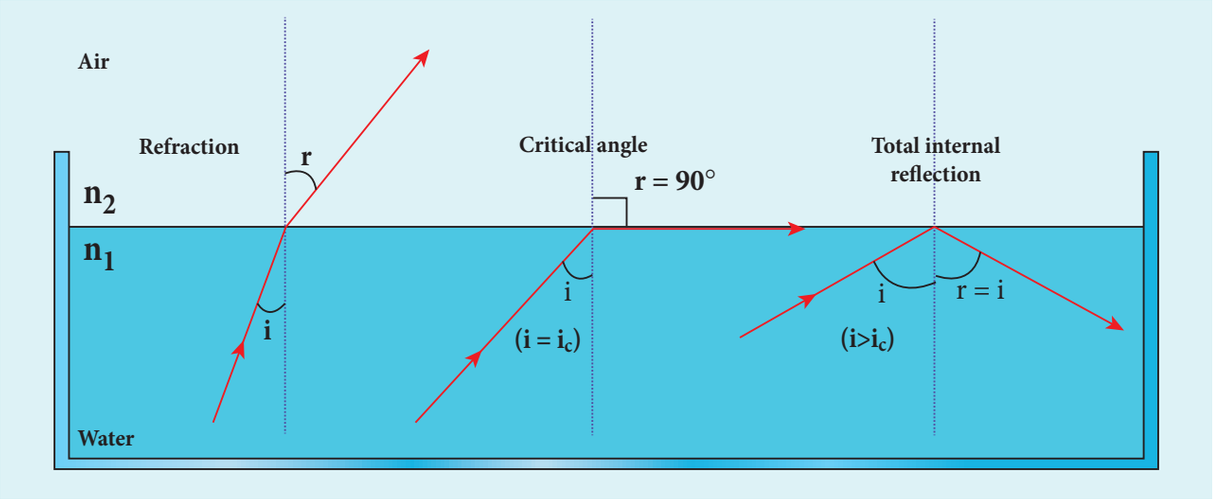


Figure 6.21 Critical angle and total internal reflection

Table 6.3 Refractive index and critical angle for different media

Material	Refractive index	Critical Angle
Ice	1.310	49.8°
Water	1.333	48.6°
Fused Quartz	1.458	43.3°
Crown Glass	1.541	40.5°
Flint Glass	1.890	31.9°
Calcite	1.658	37.0°
Strontium Titanate	2.394	24.7°
Diamond	2.417	24.4°
Rutile	2.621	22.4°

For example the refractive index of glass is about 1.5. The critical angle for glass-air interface is, $i_c = \sin^{-1}\left(\frac{1}{1.5}\right) = 41.8^{\circ}$. The refractive index of water is 1.33. The critical angle for water-air interface is, $i_c = \sin^{-1}\left(\frac{1}{1.3}\right) = 48.6^{\circ}$.

6.4.7 Effects due to total internal reflection

6.4.7.1 Glittering of diamond

Diamond appears dazzling because of the total internal reflection of light that happens inside the diamond. The refractive index of diamond is about 2.417. It is much greater than the refractive index of ordinary glass which is about only 1.5. The critical angle of diamond is about 24.4°. It is much less than that of ordinary glass. A skilled diamond cutter makes use of this larger range of angle of incidence (24.4° to 90° inside the diamond), to ensure that light entering the diamond is total internally reflected from the many cut faces before getting out as shown in Figure 6.22. This gives a sparkling effect for diamond.

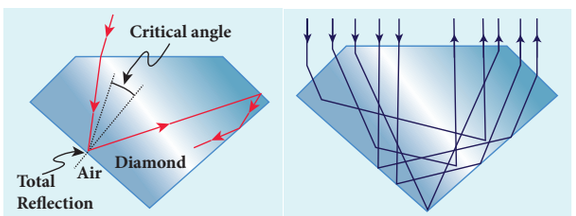


Figure 6.22 Total internal reflection in diamond

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